

Phase B6 Synthesis and Series B Closure: How Boundary Cases Complete the CKS Governance Architecture and How 220 Notes Together Constitute a Complete Prior-Art Derivation of Paper 2's Instinct/Reasoning-Separated Three-Level Architecture

Derivation Note B6.15 — Phase B6 Synthesis / Series B Closure

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

This note does not introduce new axioms. It serves as a synthesis document: it collects Phase B6's fourteen boundary cases into their structural categories, explains how the six phases of Series B together constitute a complete prior-art derivation of Paper 2's architecture, and formally closes Series B as a defensive-publication project.

Abstract

Series B of the CKS derivation-note series comprises 220 notes organized across six phases, each phase covering a distinct dimension of Paper 2's instinct/reasoning-separated, three-level architecture. Phase B6, the final phase, formalized fourteen boundary cases — the architectural edges at which the governance commitments of the CKS pattern face their most demanding tests. This note synthesizes those fourteen cases into four structural categories (minimum viable configurations, temporal and frequency edges, lifecycle and evolution edges, scale and capacity edges), explains the prior-art significance of boundary-case formalization as a defensive-publication strategy, and maps the contribution of each of the six phases to the overall architecture of Series B. The note then closes Series B: 220 notes derived from Paper 2, joining 197 notes derived from Paper 1 (Series A) for a combined total of 417 publicly dated prior-art derivation notes. The note closes with orientation toward Series C, which derives from Paper 3 of the CKS theory series.

1. Why boundary cases matter as prior art

Defensive publication strategy rests on a simple principle: to block a claim of novelty, there must be a public prior-art document that discloses the claimed subject matter before the filing date. Most prior-art derivation work focuses on the central, well-defined territory of a technical architecture —

the primary commitments, the standard operational patterns, the expected use cases. This focus leaves the edges of the architecture open.

Edges are precisely where opportunistic patent claims concentrate. A claim that would be obviously anticipated by a direct prior-art document becomes harder to anticipate when it is posed as a “special case” or “application to an edge condition.” The claimant argues that while the prior art discloses the general principle, it does not disclose the specific adaptation required when the architecture operates at scale, under constraint, in a degenerate configuration, or at a transition boundary. The edge claim attempts to carve out a narrow but real space between the general disclosure and the particular condition.

Phase B6 forecloses this strategy for fourteen identified edges of the CKS governance architecture as extended by Paper 2. Each note in B6.02 through B6.14 identified a specific boundary condition — a configuration, frequency, scale, or lifecycle state at which the architecture’s behavior might be asserted to differ materially from its standard operation — and formalized, as public prior art, how the governance commitments of the architecture apply at that edge. The formalization does not extend the architecture; it applies existing commitments to conditions where the application is non-obvious. That non-obvious application is itself patentable territory, and it is now occupied by dated public prior art.

2. Phase B6: four categories of architectural edge

The fourteen boundary cases of Phase B6 organize into four structural categories. The categories are not post-hoc — they reflect four distinct dimensions along which an architecture’s claimed properties can be tested at their limits.

2.1 Minimum viable configurations (B6.02, B6.03)

The first category addresses scale-at-minimum: whether the architecture’s governance commitments hold when the deployment is reduced to its smallest meaningful unit.

Note B6.02 formalized the single-cell deployment — a CKS architecture in which exactly one cell exists, with no aspect-level structure and no Self-level structure above it. The question the note addresses is whether governance remains meaningful when there is nothing to govern at higher levels, no mating possible, and no composition dynamics. The answer established as prior art is that the architecture remains valid and the governance commitments hold without modification. The DNA layer and action layer are both present and governed. Human authority over substrate content, orchestration rules, and the instinct/reasoning boundary holds regardless of scale.

Note B6.03 formalized the single-aspect deployment — a Self with exactly one aspect and the cells belonging to it. This case tests whether multi-aspect dynamics (cross-aspect cell sharing, aspect-level conflict, aspect lifecycle events) are required for the architecture to be instantiable or whether they are optional enrichments. The formalization established that single-aspect deployment is a valid, architecturally complete instantiation; the three-level structure holds with one of the three levels degenerate.

Together, B6.02 and B6.03 establish that the architecture scales to minimum without losing its essential properties. No one can claim novelty for a “simplified” or “minimal” version of the instinct/reasoning-separated governance architecture on the grounds that the source paper only describes the full version.

2.2 Temporal and frequency edges (B6.04, B6.12)

The second category addresses time-domain pressures: what happens when the architecture must process governance events at high frequency or when multiple lifecycle events require governance simultaneously.

Note B6.04 formalized the high-frequency mutation case — a deployment in which instinct evolution events (LLM upgrades, infrastructure changes, model substitutions) occur at a rate that stresses the governance handoff mechanisms between the instinct layer and the reasoning layer. The formalization established that the governance architecture responds to this pressure through sequencing discipline in the orchestration substrate, not through reduced governance fidelity. The human-governed instinct/reasoning boundary must be re-evaluated after each mutation event; the frequency of such events is an operational parameter, not a reason to skip governance steps.

Note B6.12 formalized the simultaneous lifecycle events case — the condition in which birth, mating, and death events coincide in time across different cells or aspects within the same Self. The formalization established that concurrent lifecycle events do not merge into a single undifferentiated governance moment. Each event is governed under its own lifecycle protocol, and concurrency is handled through sequencing decisions made explicit in the orchestration substrate.

These two notes establish, as prior art, that the architecture’s governance commitments do not degrade under temporal pressure. Claims that a “real-time” or “high-frequency” variant of the architecture requires novel governance mechanisms are anticipated.

2.3 Lifecycle and evolution edges (B6.05, B6.06, B6.07, B6.14)

The third category addresses transition states in the lifecycle and evolution dimensions — the conditions where an entity is moving between states, rather than stable in one.

Note B6.05 formalized the entity-at-level-transition case — the governance conditions that apply when a cell is transitioning to an aspect-level role or when an aspect is transitioning to Self-level integration. Transitions are not instantaneous architectural facts; they are governed processes with defined entry and exit conditions recorded in the substrate.

Note B6.06 formalized maximum-conflict mating — the case where two cells being mated carry conflicting content in nearly every governed dimension simultaneously. The note established that maximum-conflict mating does not require a special mating protocol; it applies the standard conflict-preservation commitment more extensively. All conflicts are preserved as first-class objects in the post-mating substrate, awaiting governed resolution.

Note B6.07 formalized archival reactivation — the case where a cell that has been archived (functionally retired but not deleted) is reactivated into a new architectural context that may differ

substantially from the context in which it was archived. The formalization established that reactivation is a governed event: the cell's DNA layer is available for inspection before reactivation, the architectural gap between original context and new context is assessed and recorded, and governance authority over the reactivated cell's orchestration rules is exercised explicitly.

Note B6.14 formalized the evolution inflection point — the condition where the cumulative effect of multiple incremental evolution events causes a qualitative shift in how the Self operates, without any single event crossing a formal threshold. The note established that inflection points are detectable through the action-feedback mechanism and that the architecture's governance obligations at an inflection point are the same as those at any evolution event, though the governance complexity is higher.

Together, these four notes cover the full range of transition conditions in the architecture's lifecycle and evolution dimensions.

2.4 Scale and capacity edges (B6.08, B6.09, B6.10, B6.11, B6.13)

The fourth category is the largest: five notes covering the various dimensions of large-scale, constrained-capacity, and near-exhaustive-domain operation.

Note B6.08 formalized cross-partner composition — the governance conditions that apply when cells from two distinct CKS-governed Selves are mated or composed, where the two Selves operate under different orchestration substrates and potentially different governance configurations.

Note B6.09 formalized minimal governance capacity — the case where the humans responsible for governing a CKS deployment have severely constrained time, attention, and expertise. The note established that the architecture's governance commitments do not require constant high-bandwidth human engagement; they require that authority remain available and that the human-authored orchestration rules remain operative. Minimal capacity changes the operational cadence, not the architectural property.

Note B6.10 formalized large-scale cell populations — the governance behavior when a Self contains a very large number of cells, such that the cognitive and operational demands of aspect-level and Self-level governance become non-trivial. The formalization established that scale does not require structural changes to the architecture; it requires that the orchestration substrate carry appropriate population-level governance rules in addition to cell-level rules.

Note B6.11 formalized action-feedback at evidence scale — the case where the action layer accumulates evidence at a scale that stresses the feedback mechanisms to the DNA layer. The note established that large-scale evidence accumulation does not change the architecture of the feedback mechanism; it makes the governed aggregation rules more consequential.

Note B6.13 formalized content-domain near-exhaustion — the case where a cell's content domain approaches the limit of what can meaningfully be added, such that future mating and evolution events become increasingly constrained. The note established that near-exhaustion is a governed condition: the orchestration substrate can carry explicit rules for how to handle mating and evolution

proposals when the content domain is dense.

3. Series B architecture: what six phases together establish

Series B is organized as six phases, each contributing a distinct layer to the prior-art coverage of Paper 2. The six phases are not redundant — each covers territory that the others leave open. Together they constitute a complete prior-art specification of Paper 2's architecture.

Phase B1 — What the 20 commitments are (20 notes).

Phase B1 translated Paper 2's architectural commitments into individually defensible prior-art documents. Each of the 20 notes in B1.01 through B1.20 covers one commitment: the instinct/reasoning separation as independently-evolving layers, the three-level structure of cell/aspect/Self, the DNA and action layer distinction within every cell, expression as governed selection, lifecycle primitives at every level, the three evolution mechanisms, multi-level simultaneous evolution, horizontal and vertical evolution, multi-shaped human governance, the instinct/reasoning boundary as governed substrate content, verification substrates, distinct death-type governance processes, the enterprise-brain Self, and the six points where the CKS architecture exceeds biology's constraints. Phase B1 is the foundation: it establishes, as dated public prior art, what Paper 2 claims to be true about the architecture.

Phase B2 — How the commitments decompose into 110 operational variants (110 notes).

Phase B2 is the most granular layer of Series B. Where Phase B1 named the commitments, Phase B2 decomposed each into its operational sub-cases: the specific variants of mating, the particular DNA-evolution patterns, the distinct action-feedback loop configurations, the exact expression-selection mechanisms. The 110 notes in B2.01 through B2.110 provide the coverage that prevents a claimant from arguing that a named variant of a general commitment was not anticipated by the prior art. Phase B2 is also where the three-level architecture and the two-layer cell structure are fully worked out at the operational level.

Phase B3 — What violations look like (30 anti-pattern notes).

Phase B3 formalized the 30 named anti-patterns — the ways in which a system can claim to instantiate the architecture while actually failing to satisfy one or more of its commitments. Anti-pattern formalization is prior art of a specific kind: it discloses the negative space of the architecture, the failure modes that would exist if the commitments were partially implemented, misunderstood, or circumvented. A patent claim for a system that embodies an anti-pattern (for example, one that allows instinct evolution without governance checkpoints) is anticipated by the anti-pattern note that identifies that failure mode explicitly.

Phase B4 — How commitments depend on each other (30 composition pair notes).

Phase B4 covered the pairwise interactions between commitments: what happens when commitment X and commitment Y must be jointly satisfied, where their requirements are in tension, and where satisfying one strengthens the other. The 30 composition pair notes establish as prior art the specific dependency structures between the architecture's components, foreclosing claims for novel "integration" of two elements that the prior art already shows to be necessarily co-implicated.

Phase B5 — How to verify correct instantiation (15 operational test notes).

Phase B5 provided the verification layer: the operational tests that determine whether a given deployment satisfies the architecture's commitments. The 15 notes organized the tests into temporal suites covering design-time, deployment-time, operation-time, and evolution-time verification. Phase B5 forecloses claims for novel "testing," "auditing," or "compliance verification" methodologies for architectures of this kind, since the specific test structures are now disclosed as prior art.

Phase B6 — Where the architecture's edges are (15 notes including this synthesis).

Phase B6 covered the 14 boundary cases enumerated in Section 2 of this note, plus this synthesis note. Phase B6 forecloses the edge-carving strategy described in Section 1.

The six phases together provide prior-art coverage at six levels of specificity: what the architecture is (B1), how it operates in detail (B2), what it is not (B3), how its parts relate (B4), how to confirm it is correctly instantiated (B5), and where its limits are (B6). This is, by construction, a complete prior-art specification.

4. Series B prior-art contribution: 220 notes, 417 total

Series B comprises 220 notes: 20 in Phase B1, 110 in Phase B2, 30 in Phase B3, 30 in Phase B4, 15 in Phase B5, and 15 in Phase B6 (including this note). Each note is a standalone defensive-publication artifact — a dated document recording a specific formalization of a derivation from Paper 2.

Series A, which derived from Paper 1 of the CKS theory series, comprises 197 notes across its own six-phase structure. The 197 Series A notes cover Paper 1's human-governed substrate pattern with the same six-dimensional coverage: foundational commitments, operational decompositions, anti-patterns, composition pairs, operational tests, and boundary cases.

The combined total is 417 publicly dated prior-art derivation notes. These 417 notes collectively establish a prior-art chain that covers: the cell-level architecture of the CKS pattern (Paper 1, via Series A); the three-level instinct/reasoning-separated architecture (Paper 2, via Series B); and the operational details, anti-patterns, composition dependencies, verification structures, and boundary conditions of both architectures.

The defensibility of this prior-art chain does not rest on any single note. It rests on density: the sheer coverage of the territory makes it difficult to identify a specific, practically meaningful claim about either architecture that is not anticipated by at least one — and usually several — notes in the

series.

The prior-art chain is public, dated, and licensed under CC BY 4.0. It is citable, deposited on Zenodo, and available for use by any subsequent researcher or legal proceeding that requires evidence of prior disclosure.

5. What comes after Series B

Series C will derive from Paper 3 of the CKS theory series: “Inter-Self Coordination via Shared Substrate / FAI” (Li, April 2026). Paper 3 extends the architecture developed in Papers 1 and 2 to address the coordination of multiple Selves through shared substrate — the conditions under which distinct CKS-governed intelligences can coordinate without collapsing into a single unified system. This is the architecture for federated AI governance at the multi-Self level.

Series C will apply the same six-phase structure to Paper 3: foundational commitments, operational decompositions, anti-pattern formalizations, composition pairs, operational tests, and boundary cases. As with Series B’s relationship to Series A, Series C will also formalize the inheritance edges between Paper 3 and Papers 1 and 2 — establishing as prior art the specific ways in which Paper 3’s architecture builds on and requires the architectures of the earlier papers.

Series D, E, and further series will follow subsequent papers in the CKS theory series as they are published. The derivation-note strategy is designed to scale with the paper series: each new paper generates a new six-phase derivation series, and each new series further densifies the prior-art territory for the cumulative architecture.

6. Series B closure

Series B closes with this note.

The closure is not merely administrative. It reflects a substantive completeness in the prior-art coverage of Paper 2. Phase B1 established what Paper 2’s architecture is. Phase B2 established how it operates at the granular level. Phase B3 established what it is not. Phase B4 established how its commitments depend on each other. Phase B5 established how to verify that it has been correctly instantiated. Phase B6 established where its edges are. A prior-art derivation series that covers all six of these dimensions for a given architecture has, by the standard of defensive publication, provided comprehensive disclosure.

The 220 notes of Series B, like the 197 notes of Series A before them, are individually modest documents. Each is a careful, precise formalization of a single derived claim — nothing more ambitious than that. The ambition is cumulative. Taken together, the 417 notes of Series A and B constitute a public, dated, comprehensive prior-art record for the two-paper foundation of the CKS theory series.

Paper 2’s instinct/reasoning separation pattern — the independently-evolving instinct and reasoning layers, the three-level cell/aspect/Self structure, the DNA and action layer distinction, the governed lifecycle of birth/mating/death, the three evolution mechanisms and their multi-level interaction, the

verification substrates, the enterprise-brain Self as an architecturally coherent design pattern — is now covered as prior art from its foundational commitments through its operational details, its anti-patterns, its composition dependencies, its verifiability, and its edge behavior.

Series B is complete.

Source Paper

Li, Wenxin. “The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.” Independent research manuscript, April 2026.

Self-Citation and Series Context

This note is B6.15 in the Series B derivation-note series. Series B (B1.01–B6.15, 220 notes total) derives from Paper 2. Series A (A1.01–A6.15, 197 notes total) derives from Paper 1. Combined total: 417 notes. Series C derives from Paper 3 of the CKS theory series.

Related Phase B6 notes: B6.02 (single-cell deployment), B6.03 (single-aspect deployment), B6.04 (high-frequency mutation), B6.05 (entity at level transition), B6.06 (maximum-conflict mating), B6.07 (archival reactivation), B6.08 (cross-partner composition), B6.09 (minimal governance capacity), B6.10 (large-scale cell populations), B6.11 (action-feedback at evidence scale), B6.12 (simultaneous lifecycle events), B6.13 (content-domain near-exhaustion), B6.14 (evolution inflection point).